

Thermal Performance tests on Four Radiators

Multi Column types

Carried out for
Tianjin Jiuding Yangguang HVAC Co., Ltd.

Report 104719/1

Compiled by Tomas Webster

23 March 2023



This page is intentionally left blank

Thermal Performance tests on Four Radiators

Multi Column types

Carried out for: Tianjin Jiuding Yangguang HVAC Co., Ltd.
No. 9 Wuwei Road
Lutai Ninghe Tianjin City
China

Contract: Report 104719/1

Issued by: BSRIA Limited
Old Bracknell Lane West
Bracknell
Berkshire
RG12 7AH
UK

Telephone: +44 (0)1344 465600

Fax: +44 (0)1344 465626

Email: bsria@bsria.co.uk

Website: www.bsria.com/uk/

QUALITY ASSURANCE

Issue	Date	Compiled by:	Approved by:	Signature
-------	------	--------------	--------------	-----------

Final	23-Mar-2023	Tomas Webster	Mark Roper	
-------	-------------	---------------	------------	--

Junior Engineer

Head of
Laboratory



1053

DISCLAIMER

This Document must not be reproduced except in full without the written approval of an executive director of BSRIA. It is only intended to be used within the context described in the text.

This Document has been prepared by BSRIA Limited, with reasonable skill, care and diligence in accordance with BSRIA's Quality Assurance and within the scope of our Terms and Conditions of Business.

This Document is confidential to the client and we accept no responsibility of whatsoever nature to third parties to whom this report, or any part thereof, is made known. Any such party relies on the Document at its own risk.

SUMMARY

Thermal performance tests were carried out on four radiators in accordance with the procedures contained in BS EN 442-2:2014, using low pressure hot water in the BSRIA radiator test room, described within Section 3 of this report. The work was requested by Tianjin Jiuding Yangguang HVAC Co., Ltd. and was carried out during the period 16 February to 10 March 2023.

A summary of the thermal performance is shown below.

Test reference	Model Reference	Width	Height	Columns	Output W at 30ΔT	Output W at 50ΔT	Factor 'Km'	Exponent 'n'
104719A1AF	Steel 2	465	1800	10 x 2	666	1327	6.8095	1.3477
104719A2TW	5025	475	1800	8 x 2	688	1329	8.5601	1.2897
104719A3TW	Steel 3	465	1800	10 x 3	888	1687	12.3474	1.2570
104719A4AF	6812	619	1800	8 x 2	900	1615	18.3189	1.1450

Full test results are contained within section 5 and 6 of this report.

CONTENTS

1	INTRODUCTION.....	6
2	ITEMS RECEIVED FOR TEST	7
3	TEST FACILITY	8
4	INSTRUMENTATION	9
5	THERMAL RESULTS.....	10
6	PRESSURE TESTS	15

APPENDICES

APPENDIX A:	Product information	16
APPENDIX B:	Manufacturers product information	21

1 INTRODUCTION

Thermal performance tests were carried out on four radiators in accordance with the procedures contained in BS EN 442-2:2014, using low pressure hot water in the BSRIA radiator test room, described within Section 3 of this report. The work was requested by Tianjin Jiuding Yangguang HVAC Co., Ltd. and was carried out during the period 16 February to 10 March 2023.

Three samples were received on 7 February 2023. One more sample was received on 9 February 2023. One sample had received minor cosmetic damage, but this was assessed to not be significant for thermal output. All other samples were received in good condition.

This report relates to the samples tested and no others.

Comments and opinions are outside the scope of UKAS accreditation.

2 ITEMS RECEIVED FOR TEST

The test samples consisted of two Multi Column types from Tianjin Jiuding Yangguang HVAC Co., Ltd..

The samples were designated by the manufacturer as follows:

Steel 2

Steel 3

5025

6812

The samples were four tapping multi column radiators. Radiators were plumbed top and bottom same end (T.B.S.E.) with the internal blocker removed.

The radiators were installed as standard for BS EN 442 but using the supplied mountings for spacing from the test chamber wall.

Further details of the test samples can be found in Appendix A, with manufacturer's drawings included in Appendix B to enable the sample types, sizes, test pressures and finish to be verified.

3 TEST FACILITY

The test facility consists of a test room 4.0m (l) x 4.0m (w) x 3.0m (h), which is constructed to the requirements contained in BS EN 442-2:2014, i.e., five water cooled surfaces and one insulated surface against which the test radiator is installed.

When steady state conditions are achieved the appliance output is determined from measurements of the water flow rate and inlet/outlet water temperature difference.

A standard test consists of three test points. For all appliances a first test is carried out with water supply temperature that produces a mean water temperature of 70°C with the water flow rate such that for radiators the inlet/outlet water temperature difference is 10°C. Two further tests are carried out at the water flow rate established in the first test but with different supply water temperatures.

For products marked¹ the temperature drop was set at 5°C to enable the flow rate to remain within the standard measurement uncertainty for the test chamber.

For all tests the enclosure air temperature is controlled to maintain 20°C at the inner room reference point, which is 0.75m from the floor in the centre of the room.

4 INSTRUMENTATION

TEST RECORD SHEET TP21/7 : TEST EQUIPMENT / INSTRUMENTS

	Instrument No.	Calibration expiry date
Weigh scales	2026, 2027	16/01/2024
Resistance thermometer (air) reference & radiant shield	329	13/09/2023
Resistance thermometers (water) reference	1512-1515	09/08/2023
Digital voltmeter 7½ digit (resistance thermometer measurement)	266	09/08/2023
Barometer	881	02/12/2023
Electronic timer within PC	526	07/12/2023
Vernier caliper (0 – 1.5m)	359	22/03/2025
Digital caliper (0 – 150mm)	2057	22/03/2023
Weigh scales (water content)	188	17/01/2024

Comments: None

Test Engineer

Tomas Webster

5 THERMAL RESULTS

Test reference	Model Reference	Width	Height	Columns	Output W at 30ΔT	Output W at 50ΔT	Factor 'Km'	Exponent 'n'
104719A1AF	Steel 2	465	1800	10 x 2	666	1327	6.8095	1.3477
104719A2TW	5025	475	1800	8 x 2	688	1329	8.5601	1.2897
104719A3TW	Steel 3	465	1800	10 x 3	888	1687	12.3474	1.2570
104719A4AF	6812	619	1800	8 x 2	900	1615	18.3189	1.1450

Results for the products are for those heights tested, enabling catalogues to be generated based on overall linear length for each height tested.

Catalogue outputs can be derived for other lengths of those heights tested by using a linear length constant of 1 (divide test output by test length for relevant height and multiply by desired length).

Test data is valid for the type, material and surface finish tested. The data may also be valid for other surface finishes with the same emissivity.

Individual test results for samples in this contract are shown below.

Date of Test: 16/02/2023
 Manufacturer: Tinajin Jiuding Yangguang HVAC Co., Ltd
 Model Reference: Steel 2
 Test Reference Number: 104719A1AF
 Type of Heater: Multi Column
 Surface Finish: White Powder Coat
 Pipework Connections: T.B.S.E.

HEATER DIMENSIONS

Overall Height: (mm) 1800
 Overall Length: (mm) 460
 Overall Depth: (mm) 70
 Convactor Height: (mm) N/A
 Convactor Depth: (mm) N/A
 Height above floor: (mm) 110
 Distance from wall: (mm) 45

Radiated heat factor Sk 0.30
 Barometer exponent np 0.50

MEAN TEST VALUES	TEST 1	TEST 2	TEST 3
Room air temperature 0.75m: (°C)	19.6	19.8	19.6
Flow rate (g/s)	29.0	28.8	29.3
Flow enthalpy (J/g)	313.9	219.5	360.3
Return enthalpy (J/g)	268.4	196.5	303.0
Flow temperature (°C)	75.0	52.4	86.1
Return temperature (°C)	64.1	46.9	72.4
Output (W)	1323.1	662.0	1680.5
Mean water temperature (°C)	69.6	49.7	79.2
Temperature difference (°C)	49.9	29.9	59.6
Barometric pressure (mbar)	1011.2	1010.6	1010.3
Corrected output (W)	1323.9	662.5	1682.1
Estimated output (W)	1323.9	662.5	1682.2

PERFORMANCE EQUATION

$$\text{Output (W)} = K_M (\text{mean water temperature} - \text{room air temperature})^n$$

From test results $K_M = 6.8095$
 $n = 1.3477$

VARIATION OF OUTPUT WITH TEMPERATURE DIFFERENCE

TEMPERATURE DIFFERENCE °C	HEAT OUTPUT W
20	386
30	666
40	982
50	1327
60	1696
70	2088

Date of Test: 16/02/2023
 Manufacturer: Tinajin Jiuding Yangguang HVAC Co., Ltd
 Model Reference: Steel 2
 Test Reference Number: 104719A2TW
 Type of Heater: Multi Column
 Surface Finish: White Powder Coat
 Pipework Connections: T.B.S.E.

HEATER DIMENSIONS

Overall Height: (mm) 1800
 Overall Length: (mm) 460
 Overall Depth: (mm) 70
 Convactor Height: (mm) N/A
 Convactor Depth: (mm) N/A
 Height above floor: (mm) 110
 Distance from wall: (mm) 25

Radiated heat factor Sk 0.30
 Barometer exponent np 0.50

MEAN TEST VALUES	TEST 1	TEST 2	TEST 3
Room air temperature 0.75m: (°C)	19.7	19.6	19.5
Flow rate (g/s)	32.1	32.2	32.0
Flow enthalpy (J/g)	314.5	219.7	360.1
Return enthalpy (J/g)	272.8	198.1	307.2
Flow temperature (°C)	75.1	52.5	86.0
Return temperature (°C)	65.2	47.3	73.4
Output (W)	1336.5	696.6	1696.0
Mean water temperature (°C)	70.2	49.9	79.7
Temperature difference (°C)	50.5	30.3	60.2
Barometric pressure (mbar)	1011.0	1011.7	1011.6
Corrected output (W)	1337.4	697.0	1696.9
Estimated output (W)	1346.0	695.8	1688.8

PERFORMANCE EQUATION

$$\text{Output (W)} = K_M (\text{mean water temperature} - \text{room air temperature})^n$$

From test results $K_M = 8.5601$
 $n = 1.2897$

VARIATION OF OUTPUT WITH TEMPERATURE DIFFERENCE

TEMPERATURE DIFFERENCE °C	HEAT OUTPUT W
20	408
30	688
40	997
50	1329
60	1682
70	2051

Date of Test: 10/03/2023
 Manufacturer: Tinajin Jiuding Yangguang HVAC Co., Ltd
 Model Reference: Steel 3
 Test Reference Number: 104719A3TWR1
 Type of Heater: Multi Column
 Surface Finish: White Powder Coat
 Pipework Connections: T.B.S.E.

HEATER DIMENSIONS

Overall Height: (mm) 1800
 Overall Length: (mm) 470
 Overall Depth: (mm) 98
 Convactor Height: (mm) N/A
 Convactor Depth: (mm) N/A
 Height above floor: (mm) 110
 Distance from wall: (mm) 35

Radiated heat factor Sk 0.30
 Barometer exponent np 0.50

MEAN TEST VALUES	TEST 1	TEST 2	TEST 3
Room air temperature 0.75m: (°C)	20.5	20.0	19.8
Flow rate (g/s)	37.2	37.1	37.3
Flow enthalpy (J/g)	319.1	220.0	364.2
Return enthalpy (J/g)	274.5	196.3	306.5
Flow temperature (°C)	76.2	52.6	87.0
Return temperature (°C)	65.6	46.9	73.2
Output (W)	1657.1	879.7	2150.9
Mean water temperature (°C)	70.9	49.7	80.1
Temperature difference (°C)	50.4	29.8	60.3
Barometric pressure (mbar)	998.8	998.9	982.6
Corrected output (W)	1665.3	884.1	2174.0
Estimated output (W)	1704.9	878.8	2136.2

PERFORMANCE EQUATION

$$\text{Output (W)} = K_M (\text{mean water temperature} - \text{room air temperature})^n$$

From test results $K_M = 12.3474$
 $n = 1.2570$

VARIATION OF OUTPUT WITH TEMPERATURE DIFFERENCE

TEMPERATURE DIFFERENCE °C	HEAT OUTPUT W
20	533
30	888
40	1274
50	1687
60	2121
70	2575

Date of Test: 21/02/2023
 Manufacturer: Tinajin Jiuding Yangguang HVAC Co., Ltd
 Model Reference: 6812
 Test Reference Number: 104719A4AF
 Type of Heater: Multi Column
 Surface Finish: White Powder Coat
 Pipework Connections: T.B.S.E.

HEATER DIMENSIONS

Overall Height: (mm) 1800
 Overall Length: (mm) 620
 Overall Depth: (mm) 58
 Convector Height: (mm) N/A
 Convector Depth: (mm) N/A
 Height above floor: (mm) 110
 Distance from wall: (mm) 35

Radiated heat factor Sk 0.30
 Barometer exponent np 0.50

MEAN TEST VALUES	TEST 1	TEST 2	TEST 3
Room air temperature 0.75m: (°C)	19.5	19.6	19.5
Flow rate (g/s)	39.6	39.5	39.3
Flow enthalpy (J/g)	315.1	216.0	355.5
Return enthalpy (J/g)	271.4	194.1	307.7
Flow temperature (°C)	75.3	51.6	84.9
Return temperature (°C)	64.8	46.4	73.5
Output (W)	1733.8	863.9	1881.3
Mean water temperature (°C)	70.1	49.0	79.2
Temperature difference (°C)	50.5	29.4	59.7
Barometric pressure (mbar)	1007.0	1006.7	1002.0
Corrected output (W)	1737.4	865.8	1888.5
Estimated output (W)	1634.5	878.4	1978.7

PERFORMANCE EQUATION

$$\text{Output (W)} = K_M (\text{mean water temperature} - \text{room air temperature})^n$$

From test results $K_M = 18.3189$
 $n = 1.1450$

VARIATION OF OUTPUT WITH TEMPERATURE DIFFERENCE

TEMPERATURE DIFFERENCE °C	HEAT OUTPUT W
20	566
30	900
40	1251
50	1615
60	1990
70	2374

6 PRESSURE TESTS

TEST RECORD SHEET TP21/4 : BURST TEST

BS EN 442-1 : 2014 clause 5.6 requirements

Sample heating appliances shall be subjected to a burst test at a pressure 1.69 times the maximum working pressure (See commentary below).

- The sample under test may deform but shall not rupture.
- The sample radiators shall not be less than 500mm long.

Commentary

The leak testing pressure is specified by the manufacturer (usually on the product drawing) and is at least 1.3 times the quoted maximum operating pressure. This pressure shall not be less than 520 kPa (75.4 psi). BS EN 442-1 : 2014 clause 5.4 refers. Furthermore, the burst test pressure shall be maintained for a minimum period of 2 minutes.

Test instrumentation

	BSRIA instrument number	Calibration expiry date
Standard pressure gauge	1535	28/04/2023
Stopwatch	183	20/01/2024

Test data

Test item reference number	Date of Test	Burst test pressure [1.69 x maximum operating pressure] (Bar)	Did test item hold burst pressure for a minimum of 2min without rupture Pass/Fail
104719A1AF	16/03/2023	16.9	Pass
104719A2TW	16/03/2023	16.9	Pass
104719A3TW	16/03/2023	15.21	Pass
104719A4AF	16/03/2023	5.92	Pass

TEST ENGINEER

Tomas Webster

APPENDIX A: PRODUCT INFORMATIONTEST RECORD SHEET TP21/1: TEST ITEMS

Contract number		104721/A	
Date of receipt	Test Engineer initials	Full description of test item	Test item reference number
07/03/2023	AF	White multi column radiator with two rows, 1800x465	104719A1AF
	TW	White multi column radiator with two rows, 1800x475	104719A2TW
	TW	White multi column radiator with three rows, 1800x475	104719A3TW
09/03/2023	AF	White multi column radiator with two rows, 1800x619	104719A4AF

Comments:- Test item reference numbers list all items received in the numerical order they were tested.

Test Engineer (signature)

Tomas Webster

TEST RECORD SHEET TP21/2 : PRODUCT INFORMATION

BSRIA test reference number	104719A1AF
Client	Tianjin Jiuding Yangguang HVAC Co., Ltd.
Manufacturer	Tianjin Jiuding Yangguang HVAC Co., Ltd.
Product reference number	Steel 2
Product style	Multi Column
Material of construction	Steel, powder coat white
Date of receipt	07/02/2023
Product or packaging markings	Model, Dimensions
Test start date	16/02/2023
Weight (dry) (kg)	31.7
Water content (kg)	14.28

DIMENSIONAL MEASUREMENTS

Measurement Parameter	Measured value (mm)	Manufacturer's stated value (mm)	EN 442-2 dimensional tolerance (mm or %)	Pass / Fail
Overall height	1800	1800	+6/-2	Pass
Overall length	460	465	+/-1.5%	Pass
Overall depth	70	70	+/-1.5	Pass
Convactor height	N/A			
Convactor depth	N/A			

Number of columns per panel	10 x 2
Distance installed from the wall (mm)	45
Distance between centres (mm)	1725
Panel thickness (mm)	25.13
Convactor overall length (mm)	N/A
Convactor thickness (mm)	N/A
Spot weld horizontal pitch (mm)	45.0
Additional information	T.B.S.E. connections

TEST ENGINEER (Signature)

Tomas Webster

TEST RECORD SHEET TP21/2 : PRODUCT INFORMATION

BSRIA test reference number	104719A2TW
Client	Tinajin Jiuding Yangguang HVAC Co., Ltd.
Manufacturer	Tinajin Jiuding Yangguang HVAC Co., Ltd.
Product reference number	5025
Product style	Multi Column
Material of construction	Steel, powder coat white
Date of receipt	07/02/2023
Product or packaging markings	Model, Dimensions
Test start date	17/02/2023
Weight (dry) (kg)	42.1
Water content (kg)	19.39

DIMENSIONAL MEASUREMENTS

Measurement Parameter	Measured value (mm)	Manufacturer's stated value (mm)	EN 442-2 dimensional tolerance (mm or %)	Pass / Fail
Overall height	1800	1800	+6/-2	Pass
Overall length	475	475	+/-1.5%	Pass
Overall depth	80	80	+/-1.5	Pass
Convactor height	N/A			
Convactor depth	N/A			

Number of columns per panel	8 x 2
Distance installed from the wall (mm)	25
Distance between centres (mm)	1755
Panel thickness (mm)	25.55
Convactor overall length (mm)	N/A
Convactor thickness (mm)	N/A
Spot weld horizontal pitch (mm)	58.75
Additional information	T.B.S.E. connections

TEST ENGINEER (Signature)

Tomas Webster

TEST RECORD SHEET TP21/2 : PRODUCT INFORMATION

BSRIA test reference number	104719A3TW
Client	Tinajin Jiuding Yangguang HVAC Co., Ltd.
Manufacturer	Tinajin Jiuding Yangguang HVAC Co., Ltd.
Product reference number	Steel 3
Product style	Multi Column
Material of construction	Steel, powder coat white
Date of receipt	07/02/2023
Product or packaging markings	Model, Dimensions
Test start date	10/03/2023
Weight (dry) (kg)	47.1
Water content (kg)	21.20

DIMENSIONAL MEASUREMENTS

Measurement Parameter	Measured value (mm)	Manufacturer's stated value (mm)	EN 442-2 dimensional tolerance (mm or %)	Pass / Fail
Overall height	1800	1800	+6/-2	Pass
Overall length	470	465	+/-1.5%	Pass
Overall depth	100	98	+/-1.5	Pass
Convactor height	N/A			
Convactor depth	N/A			

Number of columns per panel	10 x 3
Distance installed from the wall (mm)	28
Distance between centres (mm)	1725
Panel thickness (mm)	25.33
Convactor overall length (mm)	N/A
Convactor thickness (mm)	N/A
Spot weld horizontal pitch (mm)	45.0
Additional information	T.B.S.E. connections

TEST ENGINEER (Signature)

Tomas Webster

TEST RECORD SHEET TP21/2 : PRODUCT INFORMATION

BSRIA test reference number	104719A4AF
Client	Tianjin Jiuding Yangguang HVAC Co., Ltd.
Manufacturer	Tianjin Jiuding Yangguang HVAC Co., Ltd.
Product reference number	6812
Product style	Multi Column
Material of construction	Steel, powder coat white
Date of receipt	09/02/2023
Product or packaging markings	Model, Dimensions
Test start date	21/02/2023
Weight (dry) (kg)	53.0
Water content (kg)	15.46

DIMENSIONAL MEASUREMENTS

Measurement Parameter	Measured value (mm)	Manufacturer's stated value (mm)	EN 442-2 dimensional tolerance (mm or %)	Pass / Fail
Overall height	1800	1800	+6/-2	Pass
Overall length	620	619	+/-1.5%	Pass
Overall depth	58	60	+/-1.5	Pass
Convactor height	N/A			
Convactor depth	N/A			

Number of columns per panel	8 x 2
Distance installed from the wall (mm)	35
Distance between centres (mm)	1758
Panel thickness (mm)	12.40
Convactor overall length (mm)	N/A
Convactor thickness (mm)	N/A
Spot weld horizontal pitch (mm)	77.5
Additional information	T.B.S.E. connections

TEST ENGINEER (Signature)

Tomas Webster

APPENDIX B: MANUFACTURERS PRODUCT INFORMATION







